

Gaussian GenAI - Synthetic Market Data Generation

Joerg Kienitz, joerg.kienitz@mrig.de

Jörg Kienitz, mrig - Director Quantitative Methods

Fachbereich Mathematik, Bergische Universität Wuppertal, African Institute of Financial Markets and Risk Management, University of Cape Town,

Standard Bank Insurance and Asset Management (LibFin) - external Quant Researcher.

Abstract: We propose a method for synthetic market data modelling under the real world measure, $\mathbb{P}$. It is based on Gaussian Mixture Models (GMM) and, thus, on well known distributions and techniques. It can be considered as non-parametric since we do not make any assumptions on the data and it adapts by fitting to observed probability densities. Once numerically fitted to observed data using standard algorithms like expectation maximization, it is analytic since from the joint distribution, closed form solutions for marginals and conditionals are available in closed form. We demonstrate the power of the method by using it for time series generation. Here backfilling of data, generation of yield curves and volatility surfaces as well as generating conditioned data is considered.

Generative AI has gained a lot of attention, especially due to the capabilities of Large Language Models (LLM) and their ever increasing potential. For Financial Markets, generative methods are of fundamental importance to evaluate trading strategies, generate realistic scenarios for stress testing and risk management, impute or backfill missing data, model illiquid markets as well as forecasts realizations. Several methods for this task have been proposed. Many of them are based on architectures of deep neural networks. This assumes availability of data for training of thousands of parameters which is not the case for standard financial time series.

Introduction Properties of Financial markets data are to a large extent determined by the choice of state variables. Too many affect the performance and causes estimation issues, too few leads to miss certain risks. This is the reason why dimension reduction, resp. reducing the number of state variables, with the least possible loss of accuracy, is of paramount importance. An early and well-known example of this is the application of principal component analysis. Since the advent of statistical learning techniques many researchers have applied new methods, (conditional) Variational Autoencoders (VAE) or Generative Adversarial Networks (GAN), to this task, [9, 10, 12, 11] - yield curves, [1] - volatility surfaces, [4, 13, 14] - time series. All these methods are based on deep neural network technology. This has the downside that large amounts of training data are used to calibrate/train the underlying network architectures.

We apply the Gaussian Mixture Model (GMM), known to work well with much less data then used for training deep neural networks, already successfully used for pricing, hedging and calibration, see [8]. A GMM is based on convex combinations of (multi-dimensional) Gaussian distributions. For applying this methodology to the task of market data generation, we first show that each mean value of the constituting distributions serves as a state/factor of the model. For instance, it describes a certain shape of a yield curve or volatility surface. The different factors, the state variables, are weighted due to the weights obtained by the convex combination. We call these factors *Gaussian Principal Components*, (GPC). They allow for an interpretation of the generated data as a random combination of the GPC. This is analogous to the eigenvalue/eigenvector interpretation of principal components. Another aspect already mentioned is, that the method does rely on a descent amount of data due to the size of the parameter space which is also reducing in size if the modelled quantities exhibit a large amount of co-movement.

For the suggested method we observe the following advantages over existing methods:

- Synthetic data generation relies on well known distributions only.
- Objective criterion for backfilling and imputation.
- The amount of training data for learning is descent and much less data is needed for the learning than necessary inferring the parameters for network based methods.
- Interpretability due to the *Gaussian Principle Components* - the means of the mixture distributions.
- Covering non-stationary cases and preserving autocorrelation for time series modelling.
- Applications to high-dimensional cases is accomodated.
- The joint, the marginals and conditional distributions are analytically available.

The key idea is to approximate the probability density of observed market data structures. With GMM, this is possible, see [2]. Once having determined the number of components, the parameters (weights, mean values and covariances) are inferred. After that calibration or learning, using the model for data generation is possible. We present the generation algorithm below. Another important aspect is that marginals and conditional distributions are available in closed form and that we deal with probability distributions. Different to methods like (conditional) VAE or GAN, the variable we condition on, is part of the model itself differently to the case of conditional VAE or GAN where the condition is just an additional variable but it does not describe a joint probability distribution.

We demonstrate that the method is capable of standard tasks including the approximation of distribution resulting from a given time series of a single asset or approximating the distribution of multiple asset's joint distribution. With the analytic representation of the conditional distribution, it is possible to generate new data given realization with respect to other underlying variables. That allows for conditional backfilling, imputation and forecasting but also modelling illiquid market data. Possible examples include generating time series for newly listed stocks on an exchange, new benchmark interest rates, adding missing values for illiquid tenors for volatility surfaces or modelling the full surface based on shapes of liquid surfaces.

Since the method approximates the joint multi-dimensional distribution, it can be applied in a variety of ways. For instance, it can directly be applied to approximating time series of a full yield curve, yield curves for different markets or volatility surfaces. Furthermore, if time series data is considered where observing windows, $\{t_{T-K-n}, t_{N-K-n+1}, \dots, t_{N-K}\}$, for the data are considered the method leads to a model of the timely evolving factors. For instance, take the time interval to be 30 days and consider the time series of a yield curve. This leads to a time series of a 30 day yield curve evolution. The resulting model is the joint distribution of the timely behaviour of the states. Generation of data then leads to realizations of the curve for a 30 period. At the same time it can be used to generate the curve for the coming k days, conditioning on the last $30 - k$ days - a truly non-Markovian model.

The article is structured as follows. First, we briefly introduce the GMM and the simulation algorithm. Then, we cover standard examples for time series for from different markets and proceed by covering yield curves and volatility surfaces. The examples reach from one to multiple dimensional applications. Finally, we consider aspects such as autocorrelation or no-arbitrage relevant for the generated data. For arbitrage, we focus on *static* and *dynamic* arbitrage. Static arbitrage would be conditions that each generated curve, resp. surface is arbitrage free whereas dynamic arbitrage refers to the timely evolution of the generated objects.

GMM Let $K \in \mathbb{N}$ be a positive integer and $Z_k, k = 1, \dots, K$ be d -dimensional random variables with $Z_k \sim \mathcal{N}_d(\mu_k, \Sigma_k)$ (denoting a d -dimensional Gaussian distribution with $\mu_k \in \mathbb{R}^d$ and $\Sigma_k \in \mathbb{R}^{d \times d}$ the corresponding mean vectors and covariance matrices). Let $n_d(\cdot; \mu_k, \Sigma_k)$ be the corresponding probability density. We consider the d -variate K -component GMM, Z_{GMM} determined by the probability distribution $p_{Z_{GMM}}$:

$$p_{Z_{GMM}}(x) = \sum_{k=1}^K \pi_k n_d(x; \mu_k, \Sigma_k), \pi_k > 0, \sum_{k=1}^K \pi_k = 1.$$

π_k are called *mixing weights* and $\text{GMM}(K)$ denotes a GMM with K components. Once π_k, μ_k and $\Sigma_k, k = 1, \dots, K$, are fixed, the conditional distribution is given analytically. Let $Z = (Z_1, \dots, Z_m, Z_{m+1}, \dots, Z_{m+n})^\top$, with $m + n = d$, $Z_{GMM} \sim \mathcal{N}_d(\mu_{Z_{GMM}}, \Sigma_{Z_{GMM}})$. We denote the realizations by X and the labels by Y . We

set $Z_{GMM} = (Y, X)^\top$ with m -, resp. n -dimensional Gaussians $Y = (Z_1, \dots, Z_m)^\top$ and $X = (Z_{m+1}, \dots, Z_{m+n})^\top$ the conditionals, $Y|X$ and marginals for Y and X are available in closed form, with

$$\mu_{Z_{GMM}} = \begin{pmatrix} \mu_Y \\ \mu_X \end{pmatrix} \quad \Sigma_{Z_{GMM}} = \begin{pmatrix} \Sigma_{YY} & \Sigma_{YX} \\ \Sigma_{XY} & \Sigma_{XX} \end{pmatrix}.$$

Taking Y for instance and by denoting $p_Z(x, y)$ the conditional density with regard to Z ,

$$p_Y(y) = \int p_Z(y, x) dx \sim \mathcal{N}_m(\mu_Y, \Sigma_{YY}), \quad (1)$$

$$p_{Y|X=x}(y) = \frac{p_Z(y, x)}{\int p_Z(y, x) dx} \sim \mathcal{N}_m(\mu_{Y|X}, \Sigma_{Y|X}) \quad (2)$$

and the means, resp. variances for the conditional distribution are given by

$$\mu_{Y|X} = \mu_Y + \Sigma_{YX} \Sigma_{XX}^{-1} (X - \mu_X), \quad (3)$$

$$\Sigma_{Y|X} = \Sigma_{YY} - \Sigma_{YX} \Sigma_{XX}^{-1} \Sigma_{XY}. \quad (4)$$

Using these results for each single component $k = 1, \dots, K$ of GMM(K) we have

$$p_{k,Y|X=x}(y) = \frac{n_k(y, x)}{\int n_k(y, x) dx} \sim \mathcal{N}_m(\mu_{k,Y|X}, \Sigma_{k,Y|X})$$

and for the conditional distribution of GMM(K):

$$p_{Y|X} = \sum_{k=1}^K \tilde{\pi}_k n_{k,Y|X}, \quad (5)$$

$$\tilde{\pi}_k \sim \frac{\pi_k \mathcal{N}_m(\mu_{k,X}, \Sigma_{k,XX})}{\sum_l \pi_l \mathcal{N}_m(\mu_{l,X}, \Sigma_{l,XX})}. \quad (6)$$

For GMM(K) given, the conditional model is determined by (5) and (6). The key observation is that all conditional expectations, variances, co-variance can be calculated analytically using (3) and (4).

The parameters Let the set $\mathcal{X}$ be of size N and for $x \in \mathcal{X}$ we assume $x \in \mathbb{R}^d$. Then, $\mathcal{X}$ is represented by an $N \times d$ matrix. Assume the number of components K for GMM(K) to be given. If θ denotes the collection of all parameters π_k , μ_k and Σ_k , $k = 1, \dots, K$, θ is obtained by maximizing the (log-)likelihood

$$l(\theta) = \sum_{n=1}^N \log \left(\sum_{k=1}^K \pi_k n_d(x_n, \mu_k, \Sigma_k) \right)$$

by *Expectation-Maximization* (EM), [6]¹.

GMM Fitting with Control Variate

Input: Training set $\mathcal{X} = (x_1, \dots, x_N)$, $x_n \in \mathbb{R}^d$, with labels $\mathcal{Y} = (y_1, \dots, y_N)$, $y_n \in \mathbb{R}$. $(\mathcal{X}, \mathcal{Y})$ are joint realizations of some random variables X, Y . X are the underlying risk factors and Y is a function of X , e.g. a payoff at maturity T for Bermudan derivatives or a variance value for a Stochastic Local Volatility model.

Let $\mathcal{X}^* = (x_1^*, \dots, x_M^*)$, $x_n^* \in \mathbb{R}^d$, be a test set.

Output: Predictions $y^* = (y_1^*, \dots, y_M^*)$, $y_j^* \in \mathbb{R}$, $y_j^* \approx \mathbb{E}[Y | X = x_j^*]$. For our examples the conditional expected values of a Bermudan derivative, or the expected variance given the value of the risk factor in a Stochastic Volatility model. For in-sample performance we take $\mathcal{X}^* = \mathcal{X}$.

Parameters: The number of components K and, if we use a control variate, its realizations $\mathcal{Z} = (Z_1, \dots, Z_N)$. Including $\mathcal{Z}$ into the fitting of GMM(K), the quantities as the conditional mean, given by $\mu_Z(x) := \mathbb{E}[Z | X = x]$,

¹For other applications than the ones discussed here, different objective functions and/or optimization methods are applied. For instance, fitting volatility surfaces uses least squares objectives and constraints on forwards and to fulfill the martingale condition.

are calculated analytically. This is useful to enhance the basic GMM fitting. For generative examples the control variate can be used to model a deterministic connection to other stochastic quantities by using a standard regression model.

Calculation: The first step, the numeric step, is to fit GMM(K) to approximate the joint distribution of $\mathcal{X}$, $\mathcal{Y}$ and $\mathcal{Z}$ using the EM. The next steps being analytic, using (3) and (4). To apply the CV in order to stabilize the model we consider

$$Y^* := Y \mid X + \beta_X(Z \mid X - \mu_{Z|X}) \quad (7)$$

with β_X minimizing the variance, (7), given by

$$\beta_{X=x} := -\frac{\text{Cov}[Y, Z \mid X = x]}{\text{Var}[Z \mid X = x]}. \quad (8)$$

For $x \in \mathcal{X}$, $\mu_{Z|X=x}$ as a conditional mean, resp. $\beta_{X=x} \in \mathbb{R}^d$ as a quotient of conditional covariance and conditional variance for Gaussian distributions are calculated analytically.

β_X can be seen as a (conditional) dependence to w.r.t. Z , given the values X , that minimize the variance for Y^* . If Z is obtained as a linear combination of functions of other stochastic factors the control variate reflects deterministic conditional behaviour. This information can be used to model differential quantities. For the purpose of synthetic data generation, the *control variate* can for instance be some regressed value given known time series.

Another technique is the application of transformations, e.g. a logarithmic transform, $\log(x) + \epsilon$, $\epsilon > 0$ but small, the LambertW transform. Such methods are used to enhance stability, keep values in a certain range, e.g. staying positive and can help to reduce the number of Gaussian factors. In fact we found that for equity and foreign exchange markets the application to invariants, i.e. log-returns and applying the LambertW transform lead to the best results.

Stochastic Simulation Algorithm

Using the notation from above, the simulation from a given GMM with K components is facilitated using following algorithm:

- Compute the vector $v = (v_1, \dots, v_K)$, $v_i = \sum_{j=1}^i \pi_j$, $i = 1, \dots, K$
- Sample $u \sim \mathcal{U}(0, 1)$
- Find the mixture component according to u .

$$k := \max_i \{v_i \leq u\}$$

- Sample $x \sim \mathcal{N}_k(\mu_k, \Sigma_k)$

The algorithm accommodates conditional sampling using (3), (4) and (6) as the resulting distribution is again GMM with K components. The corresponding means, covariances and weights are calculated analytically and used in the conditional sampling. The only thing that is different is that the conditional means, covariances and weights are functions of the variable to be conditioned on.

We stress the interpretation of μ_k , $k = 1, \dots, K$ as the Gaussian Principal Components (GPC) representing the principal shapes of the state variables. Together with the weights this is analogous to eigenvectors/eigenvalues for principal component analysis.

Market Data Generation

We consider the set-up for time series generation. For our exposition we assume $n \in \mathbb{N}$ objects and $N \in \mathbb{N}$ observations. The time series matrix $\mathcal{X}_{M,N} = (x_{ij})$, $i = 1, \dots, M$, $j = 1, \dots, N$ of observations given by:

$$\mathcal{X}_{M,N} = \begin{pmatrix} x_{1,1} & x_{1,2} & \dots & x_{1,n} & x_{1,n+1} & \dots & x_{1,N} \\ x_{2,1} & x_{2,2} & \dots & x_{2,n} & x_{2,n+1} & \dots & x_{2,N} \\ \vdots & \vdots & \dots & \vdots & \vdots & \dots & \vdots \\ x_{M,1} & x_{M,2} & \dots & x_{M,n} & x_{M,n+1} & \dots & x_{M,N} \end{pmatrix}.$$

Each observation represents a tensor that is of dimension d , a value, $d = 1$, a curve, $d = k$, a surface, $d = k \times l$ or a cube $d = k \times l \times m$. The vector $x_k := (x_{k,1}, \dots, x_{k,N})^\top$ is the time series for tensor x_k and $x^l := (x_{1,l}, \dots, x_{n,l})^\top$ being the realization of the tensors at time l . For applications it is necessary to consider observed and missing values since we apply our method for backcasting and imputation. Let I be the index set, we denote by I_o the indices where values for x_i are observed and by I_m those where values are missing.

Financial time series exhibit some special behaviour which complicates the analysis:

- Non-stationarity (mean, variance, and autocorrelation of a series are not constant over time.)
- Autocorrelation (observations exhibit dependence upon prior states)
- Trend (overarching long-term change in the mean level of a time series)
- Seasonality (pattern of rises/falls in the mean level of a series that consistently occurs across time periods)

There are standard methods for handling the trend and the seasonality and removing them for further analysis. For our examples we focus on non-stationarity and autocorrelation. Considering the j -th autocovariance for a time series, $\gamma_{jt} := \mathbb{E}[(x_t - \mu_t)(x_{t-j} - \mu_{t-j})]$, then the series is weakly stationary if $\mu_t = \mu$ and $\gamma_{jt} = \gamma_j$ for all t and the j -th autocorrelation is $\tau_j := \gamma_j / \gamma_0$. It is a measure of linear association, i.e. correlation between the current time series values with its past values after controlling for the intervening observations.

Time Series

The data sets we are working with are the SOFR, ESTR and SONIA daily data for the period from 27.8.2018 to 23.8.2024 as well as GILT rates daily data from 04.01.2016 to 30.06.2022². In this case the tensor dimension is $d = 1$, $M = 3$, resp. 1 and $N = 1524$, resp. 1641, i.e. $x \in \mathbb{R}$. The training sets are $\mathcal{X}_{ois}$ and $\mathcal{X}_{gilt}$.

Remark: Instead of using the raw data, it is often more advisable to work with *market invariants*. For equity and foreign exchange that could be logarithmic returns and for fixed income absolute returns. Such invariants are more closely showing distributional and stationary behaviour than raw data. Furthermore, transforming the data to achieve a distributional shape that is closer to a Gaussian is also possible, for instance the LambertW transform is widely applied by practitioners.

Working with invariants has the advantage that the involved distributions are closer to stationary distributions.

²Analysis has also been done for stocks data constituting the Dow Jones index and Foreign Exchange data.

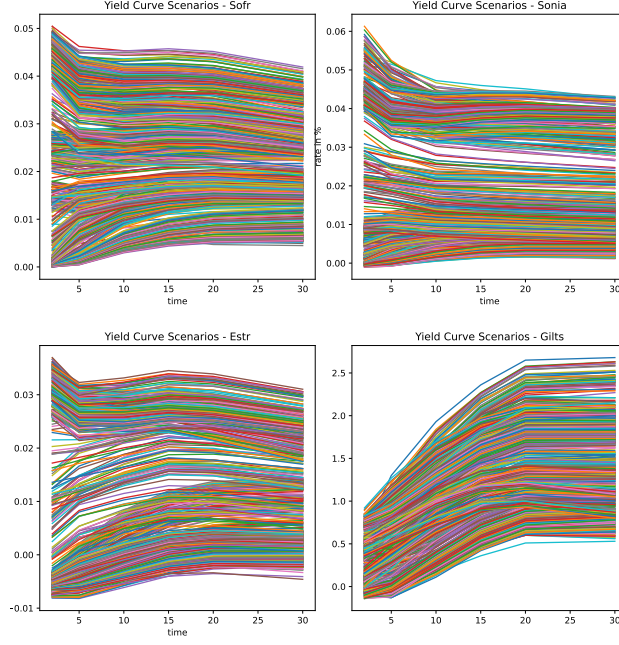


Figure 1: Curve Data for SOFR, SONIA, ESTR and GILT.

The GMM approach can fit the data very well. Figure 2 shows the marginals and the corresponding fit for the original and transformed data. Observed covariances between the rates are considered when we apply the method to yield curves.

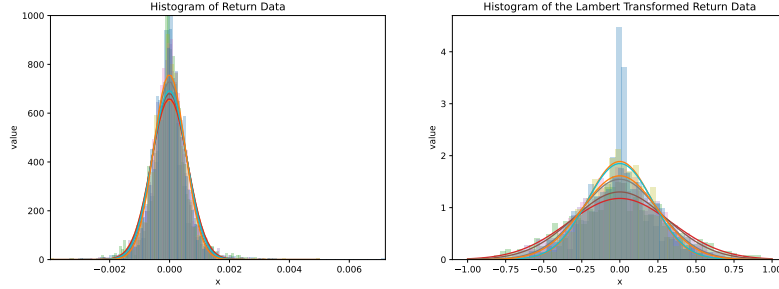


Figure 2: (left) original and (right) LambertW transformed data.

The first application we are consider, is conditional multi-variate sampling. We fit a $GMM(K)$ model to six rates from $\mathcal{X}_{gilt}$. The resulting model can then be used to infer the marginals and the conditional distributions, being $GMM(K)$ again. The simulation algorithm is applied by taking the conditional means, weights and covariances. Figure 3 displays the result for the 6 dimensional setting, and, conditioning on two, resp. three dimensions. The method not only resembles the distributions for each dimension but the realizations exhibit features observed in the time series used for training, like volatility clustering, as well.

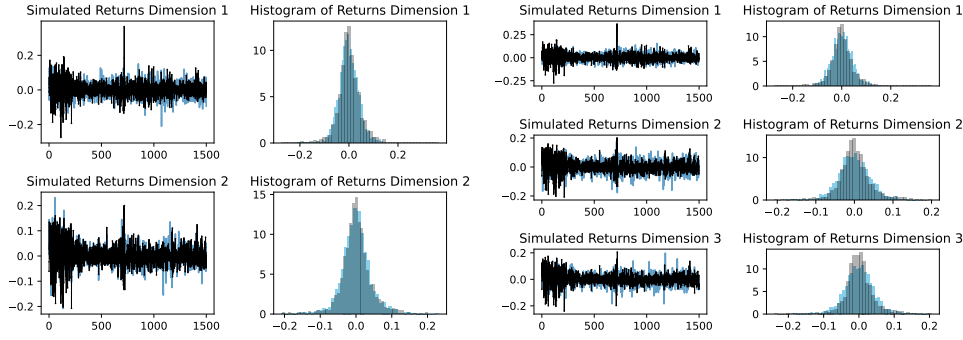


Figure 3: Simulating conditionals of GMM with 6-dim distributions conditioned on (left) 4, resp. (right) 3.

Forecasting/Backcasting/Nowcasting *Nowcasting* takes available market data and achieves the estimation of unobservable, resp. illiquid values at spot. This can for instance be a volatility surface for an illiquid asset or volatility values for illiquid tenors. I_m carries the indices for the illiquid data at certain times.

Forecasting is the process of taking available time series data and infer the distribution of the quantity of interest. This distribution is then used to compute certain figures like the mean for a point estimate or the variance to quantify the uncertainty of the point estimate. For this case $I_m = \{\}$.

Backcasting is the estimation of time series for past time intervals. In the data I_m holds the missing indices which are all indices from a given index backwards for one or more assets. It may become necessary if a new company is listed on a stock exchange or some new benchmark rate, like a risk free rate needs to be integrated into a market data set. For this application the modeler can rely on observable data, e.g. liquid indices, or foreign exchange rates that are correlated to the quantity we wish to model. Usually for this task only a sparse data set is available for fitting distributions. Here the GMM is ideally suited since it only needs relatively sparse data to train and it allows for efficiently handling conditional distributions.

The different application apply to either a standard GMM or the conditional GMM if we wish to perform conditional now-, back- or forecasting.

In the sequel we consider a challenging backcasting task also known as backfilling. To this end we consider the 2Y, 5Y, 10Y, 20Y, 25Y and the 30Y rates. For the time series we take 1500 days of history for all rates. To train the initial GMM model we use 1000 most recent data points. Then, we use the 500 remaining data points of the 2Y to 20Y rates to infer the series for the 25Y and 30Y rates. We have chosen these rates to enhance the complexity since we are extrapolating from the short to mid dated tenors to the long ones. Furthermore, we apply standard linear regression as a control variate. This is done by linear regressing the two rates using the 4 known ones for the 1000 observations.

For backcasting we proceed backwards as follows:

- Sample n times conditional on the known rates, i.e. 2Y, 5Y, 10Y and 20Y
- Fit GMM for each sample and compute the log-likelihood.
- Choose the sample s^* that minimizes the log-likelihood.
- Proceed to the next step by adding the sample s^* to the *known* data

In this way do not rely on any expert judgement or other rules of thumb but we use an *objective criterion* for the backcast, namely the minimizers of the log-likelihood when fitting a GMM model.

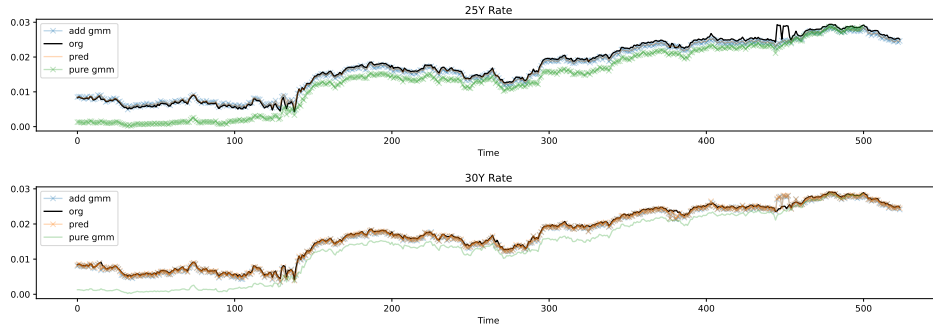


Figure 4: Backcasting series for IR data for the 25Y rate (top) and the 30Y rate (bottom) and applying the objective criterion for keeping the final realization.

Figure 4 shows the results from applying backcasting using pure regression, pure GMM and the GMM with the control variate. Since the original series is available for comparison it can be confirmed that the GMM method enhanced by the control variate performs best and nearly reproduces the path. We also wish to compare other characteristics of the generated data, namely the autocorrelation and the covariance structure.

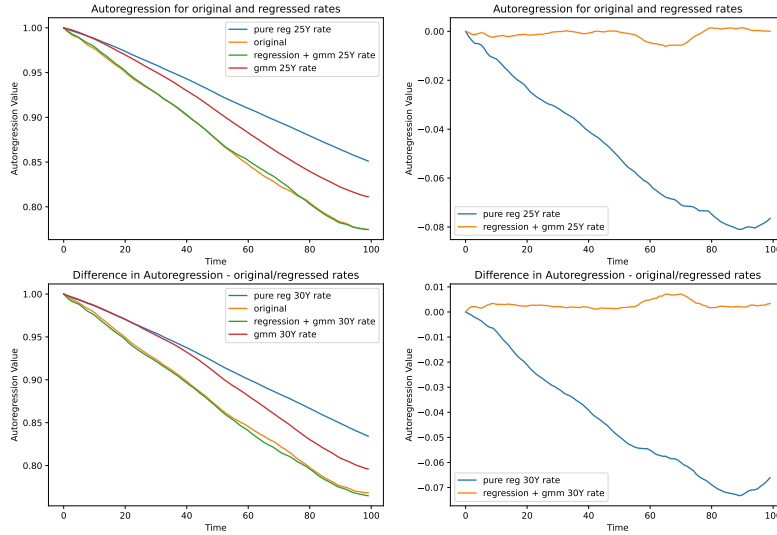


Figure 5: Backcasting series and Autocorrelation.

In Figure 5 we plot 100 day autocorrelation of the series for both rates and compare them to the actual realizations. The backcasted rates using pure regression do not resemble the autocorrelation but the ones combining regression with the GMM do, i.e. the one taking regression as a control variate approach within a GMM.

Example: Curves

Curves are an example for time series of vectors. Three options are available in this case, raw rates, continuously compounded zero rates or piecewise constant forwards and the corresponding invariants, i.e. absolute returns. For a single curve that may not make a big difference but if we consider modelling different markets at the same instant, we need to match the modeled tenors, etc. To this end using raw data may not be adequate and switching to the output of a curve building/interpolation method can be helpful. The tensor $x \in \mathbb{R}^{80}$ is of size 80 for the Gilt case and $x \in \mathbb{R}^{50}$ for the case of the other rates.

First, we train a GMM with K components, $GMM(K)$, to fit the distribution of a history of observed yield

curves for different markets. This leads to K Gaussian Principal components, μ_k , $k = 1, \dots, K$ and applying the stochastic simulation algorithm allows for simulating curves. It is important that the simulated curves not only resemble the distribution of all historic yield curve, i.e. closely resemble the covariance structures but conditioning on a curve observed today for simulation would be helpful for meaningful scenario generation. Again, fitting a GMM with K components can be used for this task. To facilitate the conditional sampling we first use a sliding windows of curves observed on n consecutive dates. The resulting data set is used to train $GMM(K)$. Since (2) represents the conditional distribution we observe that fixing $k < n$ curves results again in a $GMM(K)$ but the Gaussians are $n - k$ dimensional. Now, taking the last k curves using the stochastic simulation generates $n - k$ curves. This extension allows to cover the covariance structure not only for a given point in time but also for the evolving curves at n different time points.

Simulating a single curve After fitting a $GMM(K)$, $K = 1, 3, 5$ models to SOFR, SONIA and ESTR data at once, Figure 6 shows the resulting GPCs which are the means of the constituting distributions. Starting with a single Gaussian distribution, we observe fluctuations according to the covariance structure around this single GPC. Adding further components with the weights capture further shapes that are resembling the yield curve shapes. It is also possible to compare the different markets in terms of the shapes of the Gaussian Principal components and their corresponding weights. In the example, SOFR and SONIA are more aligned to each other than to ESTR. On the right we have plotted simulated curves from the joint distribution, by applying the simulation algorithm to this $3 \cdot 6 = 18$ dimensional setting.

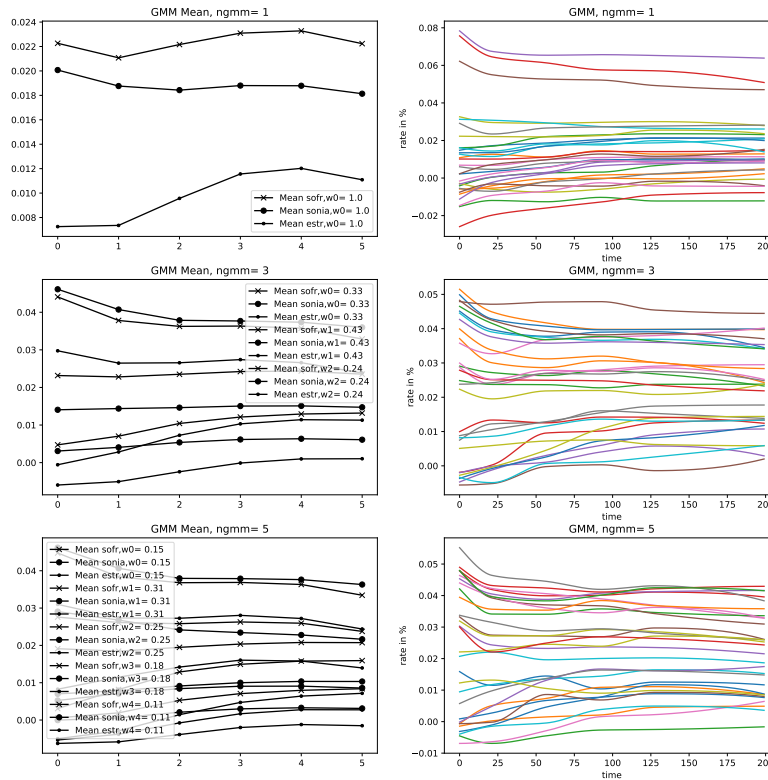


Figure 6: Gaussian Principal Components for 3 markets (left) and synthetically generated scenarios (right).

For a single currency, we tested the impact of the number of mixture components and find that recovering the variety of shapes increases. This is due to adding GPCs and, especially, adding GPCs that account for certain shapes that are not as often observed than others. Figure 7 illustrate the results. Especially, if we look at the

short end, it turns out using $K = 7$ the model starts to generate the shapes that are not that frequent in the training data. To find the optimal number of components, some cross validation has to be performed not to overfit the model.

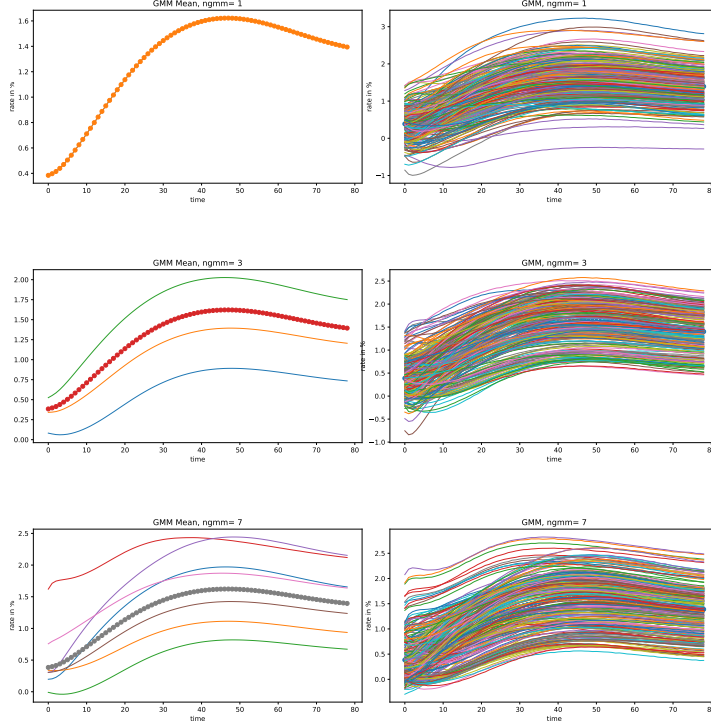


Figure 7: Gaussian Principal Components for 3 markets (left) and synthetically generated scenarios (right).

To determine the optimal number of components K , see [8]. Next, we analyze the covariance structure. To this end we again take the examples using Gilt rates, fit a $GMM(5)$ model to the data and compare the covariance structure from the training data and the synthetically generated data. Figure 8 shows that not only the marginals, displayed already in Figure 2, are resembled closely but also the covariance structure of the 80 rates. We plot 10 rates against 10 rates and the resulting covariance to the other 80 rates, starting with the first 10 in the first row. The right column in the figure gives the percentage difference which shows a maximum of about 3% at the short end for $K = 5$ and below 0.5% for $K = 7$ which is due to the fact that the variability of training data is much better captured with a more complex GMM.

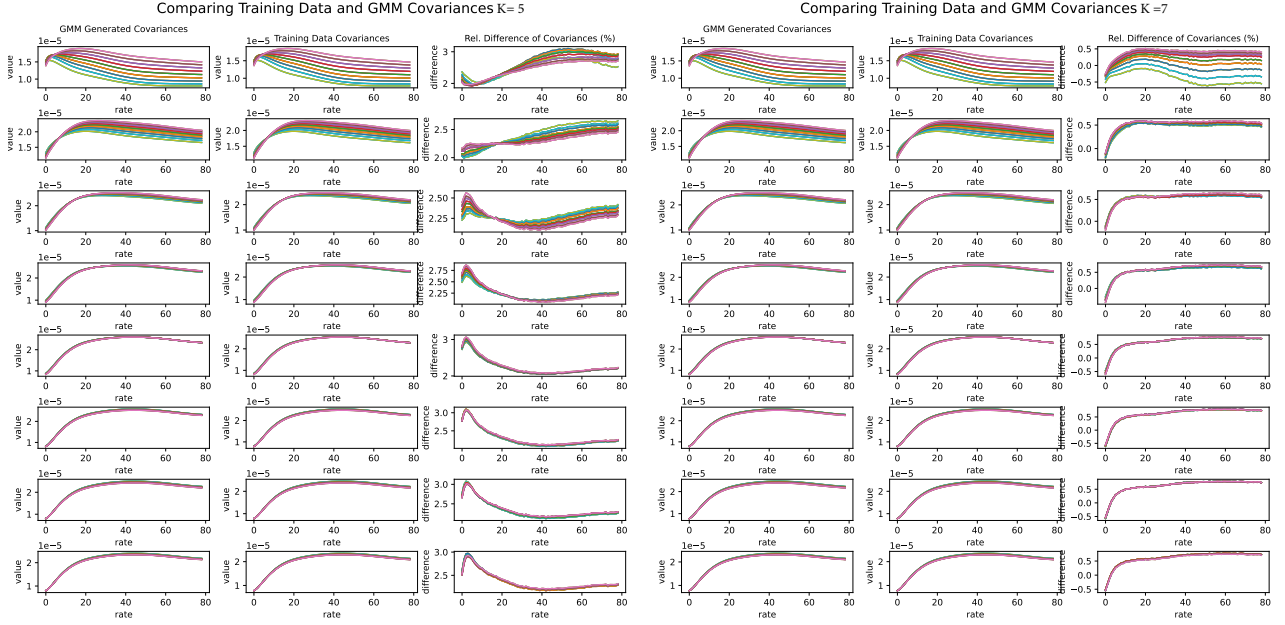


Figure 8: Covariances comparison for GMM generated curves with the training data, (left) covariances GMM generated, (mid) covariances training data, (right) differences. Left panel $K = 5$ and right $K = 7$

Simulating n curves We turn to the conditional sampling using fitted $GMM(K)$, $K = 1, 3$ to a sliding window of n curves, $n = 2, 30$. We do not have to change the fitting algorithm but just supply the new training data with the corresponding sliding windows of size n . Figure 9 gives the resulting curves ub terms of the n day evolution of the full curve. On the left just two curves are simulated for a one and a three component GMM and the variation is small which is again a realistic scenario. On the right the same simulation is done for the 30 day sliding window. Again, the consecutive variation of the generated curves is small which is realistic. It is worth to remark that with this set-up a non-Markovian dynamic is realized taking into account time and rate dependency. When we consider implied volatility surfaces below we also consider the conditional sampling.

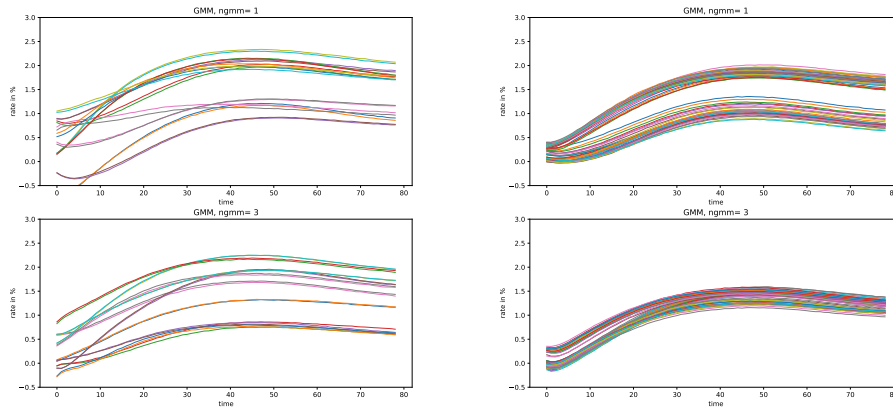


Figure 9: Sampling of two (left) and 30 day (right) sliding windows of yield curves.

With this approach it is also possible to generate a 30 day evolution under the $\mathbb{P}$ measure. Figure 10 displays generated scenarios and the GPC together with their weights.

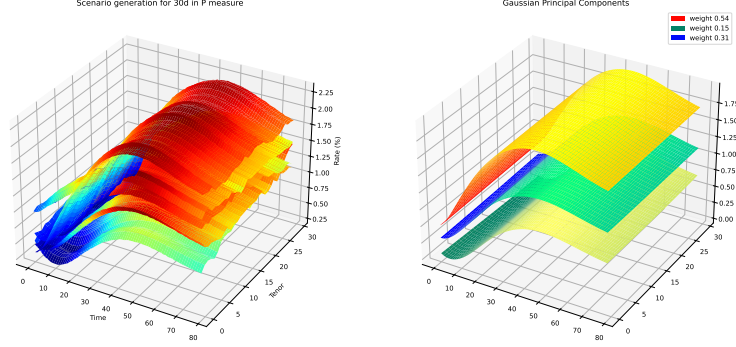


Figure 10: Sampling 5 scenarios of 30 day evolution under $\mathbb{P}$ (left) and its GPCs (right).

Example: Surfaces

Finally, we turn to synthetic generation of implied volatility surfaces. A tensor $x \in \mathbb{R}^{150}$ is of size 150 of shape 15×10 in this case. For the case where we consider one liquid and one illiquid surface it is $x \in \mathbb{R}^{150}$ with $d = 2$, thus the dimensionality is 300.

For illustrating the method, carving out the benefits in modelling and reproducibility we have used an SSVI models for generating the training data³. We use two different markets $M1$ and $M2$ represented by two different SSVI model parametrization. The implied variance surface for a given SSVI parameterization is given by:

$$w(k, \theta_t) = \frac{\theta_t}{2} \left\{ 1 + \rho \varphi(\theta_t) k + \sqrt{(\varphi(\theta_t) k + \rho)^2 + 1 - \rho^2} \right\}, \quad \rho \in (-1, 1),$$

with θ_t denotes the total at-the-money variance. Here $k := \log(K/S_0) \in \mathbb{R}$ denotes the logmoneyness. The local variance can be computed via the formula:

$$\sigma_{\text{loc}}^2(k, T) := \frac{\partial_t w(k, \theta_t)}{g(k, w(k, \theta_t))},$$

for all $k \in \mathbb{R}$ and $t \geq 0$, where the function g is fined in the paper.

Data could suggest some specific form for the increasing map $t \mapsto \theta_t$. For simplicity here, we take $\theta_t \equiv \sigma^2 t$, for some $\sigma > 0$.

The function φ is referred to as the backbone and needs to satisfy some assumptions to prevent certain types of arbitrage. We focus on two parameterizations of the SSVI model each having a different function φ , with one, resp. two additional parameters, namely

$$\varphi_1(\theta) \equiv \frac{1}{\gamma \theta} \left\{ 1 - \frac{1 - e^{-\gamma \theta}}{\gamma \theta} \right\} \text{ and } \varphi_2(\theta) \equiv \frac{\eta}{\theta^\gamma},$$

with $\gamma \geq \frac{1}{4}(1 + |\rho|)$, resp. $\gamma \in (0, \frac{1}{2}]$, $\eta > 0$. Training a GMM with different number of GPCs we generate 1000 implied volatility surfaces from each model. The training data is shown in Figure 11

³Since we provide jupyter notebooks for illustrating the method we do not wish to rely on market data that cannot be distributed. Our experiments on end of day implied volatility surfaces confirmed our findings presented here.

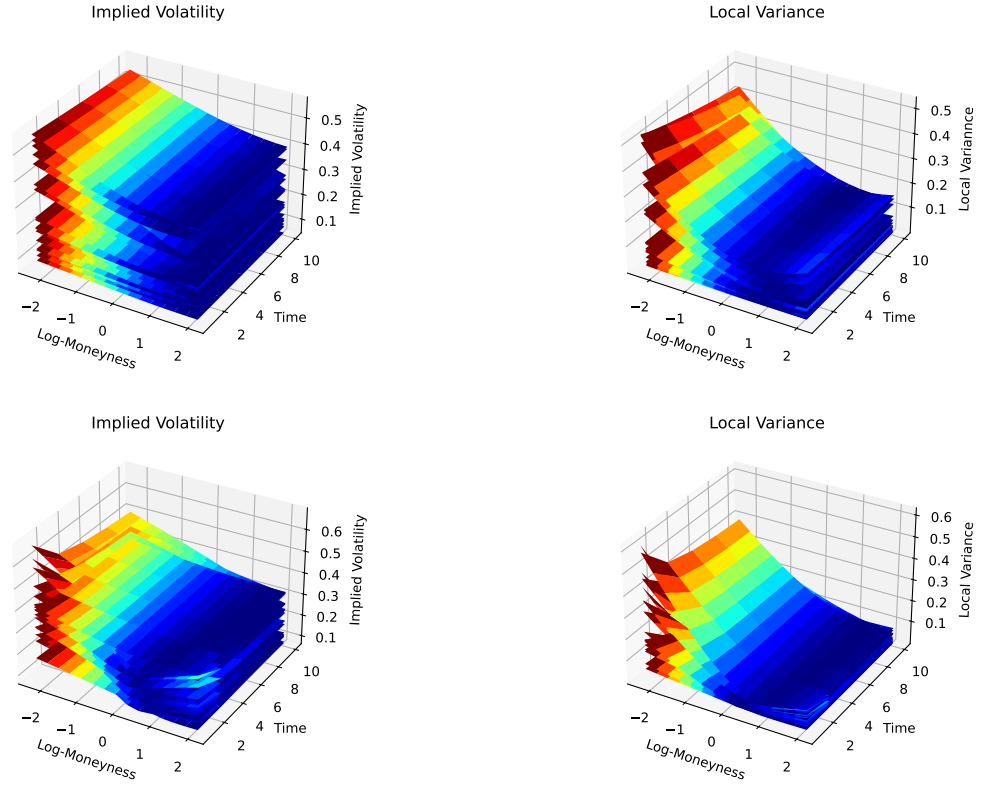


Figure 11: SSVI generated samples for φ_1 and φ_2 using different parameters θ for each sample.

Next, we fit a $GMM(3)$ to the training data, resulting in the Gaussian Principal Components shown in Figure 12. They describe the base surfaces obtained by approximating the joint distribution of all values of implied volatility for the training data.

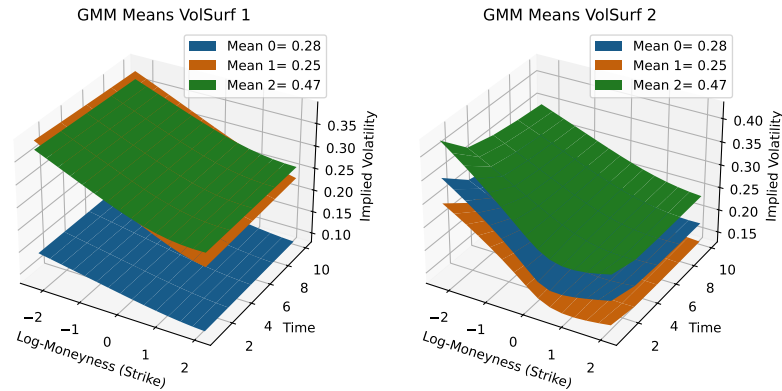


Figure 12: GPCs for the samples for φ_1 and φ_2 .

For this example, the weights for two components are nearly the same but one component accounts for 50% of the joint density. Both scenarios generated for the volatilities show, that the surfaces exhibit a very different behaviour. This is important for illustrating the versatility of the proposed method to model different implied

volatility surfaces for different markets, resp. rates. Finally, using the simulation algorithm it is possible to generate new scenarios by conditional sampling. Applying (2), we generate synthetic surfaces based on taking a realized surface for the market represented by φ_1 . That can be useful for the application to illiquid markets. Having a model for the liquid surface and using the joint distribution, it would be possible to implicitly derive a surface for the illiquid market as well. 10 samples from the conditional distribution are shown in Figure 13. On the left the implied volatility surface serves as the input to be conditioned on, market $M1$, respresented by φ_1 , whereas on the right the synthetically generated surfaces for market $M2$, represented by φ_2 , are displayed.

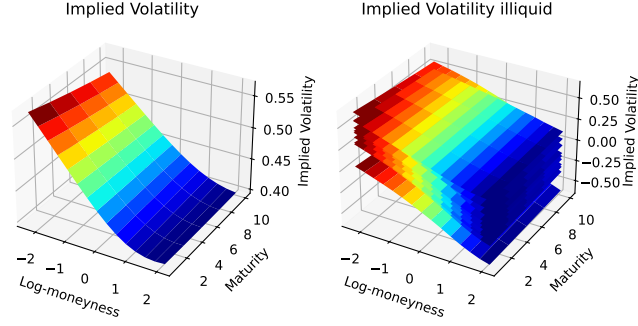


Figure 13: Conditional samples for volatility surface 2 given a specific realisation of volatility surface 1.

No-arbitrage

We consider two types of arbitrage. First, *Static Arbitrage*, is the arbitrage observed for a single financial object. For a given yield curve that means that the discount factors are monotonically decreasing. The static arbitrage for implied volatility surfaces are much harder to check and we consider the different types of arbitrage below. Second, *Dynamic Arbitrage* refers to the potential arbitrage stemming from a timely evolution of a financial object and is closely related to the risk neutral measure, $\mathbb{Q}$, dynamic. For a term structure evolution to be arbitrage free the drift has to be of special form specified by the Heath-Jarrow-Morton theorem, [11].

Static Arbitrage

We examine static arbitrage first. A method for training that accounts for reducing arbitrage can be established by penalizing corresponding outcomes when sampling. For that we include N samples and consider the ratio of observed arbitrage to N as follows:

- Start with μ_0 , Σ_0 and w_0 being the initial parameters.
- $\mu_t := \mu_0$, $\Sigma_t := \Sigma_0$, $w_t := w_0$
- Use an optimizer handling constraints for either m steps or until the objective is small enough:
 - Compute the log-likelihood using the parameters μ_t, Σ_t, w_t
 - Generate N scenarios with these parameters; count the number n of arbitrage violations.
 - Evaluate the objective function, given by the likelihood and the penalty resulting from the ratio n/N weighted by a parameter $\lambda > 0$
 - Let the optimizer update for $\mu_{t+1}, \Sigma_{t+1}, w_{t+1}$
- Keep the final parameters for parameterizing a GMM.

Another method is simpler and can be seen as approximating the trained GMM with a (nearly) arbitrage free one. For this approach we start with the approximated GMM and sample from this model only arbitrage free objects, either by repeating the sampling until we generate an arbitrage free object or by adjusting the outcomes using an arbitrage repair algorithm, see [7, 5]. Finally, we train a new GMM model to the arbitrage free initial training data augmented by the simulated arbitrage free data. If the size of the data set is large enough we may resort to only the arbitrage free data of the training data set. Figure 14 displays the differences of the methods. For $K = 7$ the approximating GMM is very close to the original GMM. There is one GPC that is not resembled by the arbitrage free data.

Both methods have clear disadvantages. The first needs a different way of training a GMM given the training data since the standard expectation maximization algorithm is not designed to incorporate constraints like arbitrage penalties and either need to incorporate such or use another optimization method. The second method does not preserve the distributional assumptions of the model!

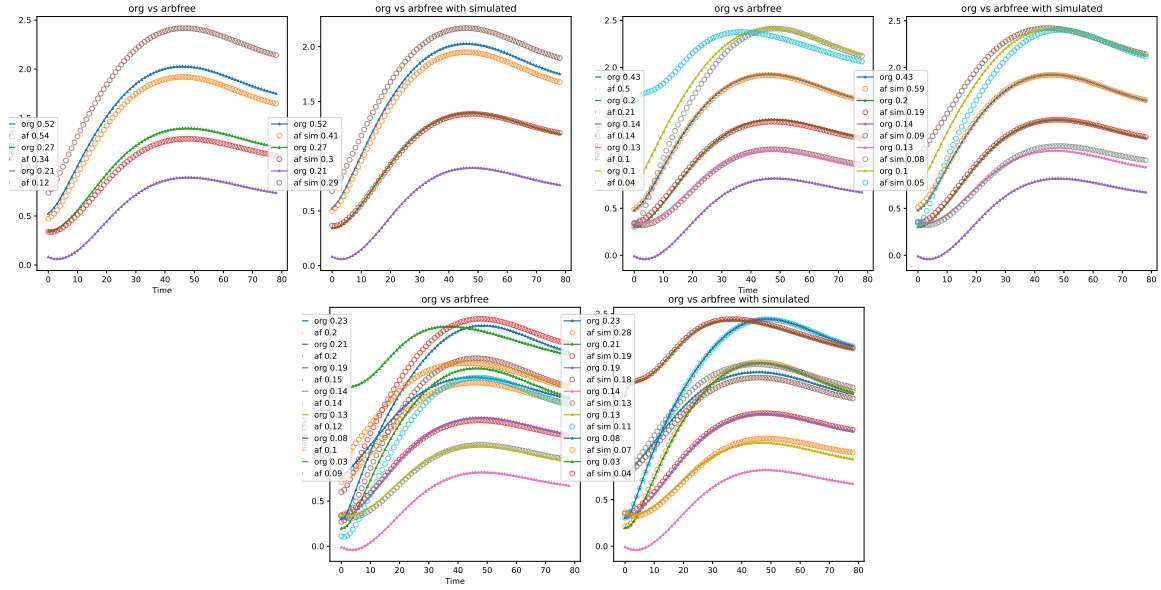


Figure 14: Comparison of the GPCs for using arbitrage free data only and arbitrage free data augmented with simulated data for $K = 3, 5, 7$.

There has been a controversy in applying statistical learning to data that need to preserve certain constraints. Either they can be handled by extending the training using constraints but it has also been observed that if the training data is chosen adequately, in our case arbitrage free, the resulting approximation is able to adapt to the data and only a small portion of the generated data does not obey the constraints. The latter is what we observed as well. The example of generating implied volatility surfaces summarizes our findings of evaluating the samples trained without any constraints. Especially, applying the training of an approximate GMM using only arbitrage free led to means, covariances and weights very close to the original one if the amount of no-arbitrage data in the training set was large.

Yield Curve - Static Arbitrage The standard assumption for no-arbitrage of a yield curve is that the discount factors are monotonically decreasing, resp. for $0 \leq t < T$ obeying the equation

$$P(0, T) = P(0, t)P(0, t, T). \quad (9)$$

Thus, there is a simple check for arbitrage available via (9). If we wish to generate only arbitrage free curves, we can apply one of the methods above. Figure 15 illustrates the situation using the approximating GMM. The reduction in variety is caused by the fact that simple no-arbitrage check - decreasing discount factors is not satisfied by the training data due to the fact that negative forwards appeared at the short end of the curve for certain markets.

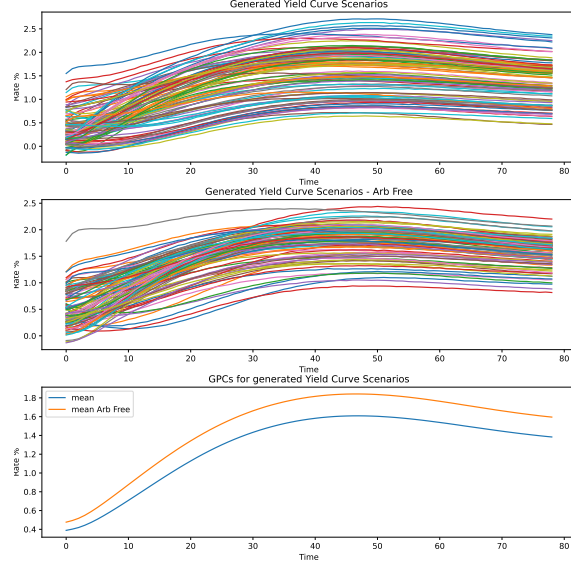


Figure 15: Yield curve scenarios generated using the learned distribution (top) and keeping only those that are arbitrage free (bottom).

Static Arbitrage - Volatility Surfaces In general, the generated surfaces need not be arbitrage free. Again, if we suppose that the input data is arbitrage free we expect that the learned distribution preserves this property to a great extent. Figure ?? confirms this expectation. From the generated implied volatility values less than 0.7% are not arbitrage free. Some arbitrage repair method can always applied to the outcome, see [7, 5].

Call price should be non-negative, (1), $C(K, T) > 0$, no vertical arbitrage, (2), $K_2 - K_1 \geq C(K_1, T) - C(K_2, T) \geq 0$ for strikes K_1, K_2 with $K_1 < K_2$ no butterfly arbitrage, (3), i.e.

$$\frac{C(K_1, T) - C(K_2, T)}{K_2 - K_1} + \frac{C(K_3, T) - C(K_2, T)}{K_3 - K_2} \geq 0, \quad K_1 < K_2 < K_3$$

No calendar arbitrage (4), i.e. $C(K, T_2) - C(K, T_1) \geq 0$ for T_1, T_2 with $T_1 < T_2$, no calendar vertical arbitrage, (5), $C(K_1, T_1) - C(K_2, T_2) \geq 0$ for $K_1 < K_2, T_1 > T_2$ and, finally, no calendar butterfly arbitrage, (6),

$$\frac{C(K_1, T_1) - C(K_2, T_1)}{K_2 - K_1} + \frac{C(K_3, T_3) - C(K_2, T_2)}{K_3 - K_2} \geq 0, \quad K_1 < K_2 < K_3, T_1 > T_2, T_3 < T_2$$

[5] proved that when all sets of call prices satisfies these conditions, then there is no static arbitrage and that these constraints can conveniently be represented with a matrix inequality which we used to analyze the arbitrage possibilities of our generation method. Finally, they provide an arbitrage repair method. Using either [5] or [7] can be used in case the generation produces non-arbitrageable surfaces. Figure 16 shows two different ways of correcting the arbitrage in a given surface and displays the resulting surfaces and the differences between the generated one and the corrected ones.

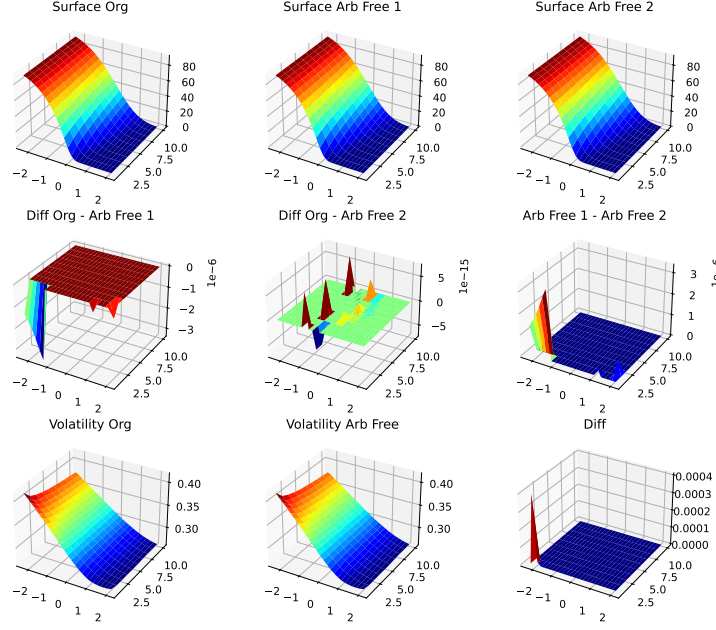


Figure 16: Synthetically generated implied volatility surface and the ones obtained after applying the arbitrage repair, resp. arbitrage free interpolation.

A final remark on arbitrage is that even if the surface generated is arbitrageable this may not be executed due to bid/ask prices. The generated surface is a mid surface.

Applying the GMM based synthetic market data generation, we have run several scenarios and recorded the type of arbitrages and the counted their appearance. In particular we consider 100 batches of simulating 100 surfaces with 250 strike, maturity pairs. This makes 25.000 implied volatility values. The arbitrage check on average leads to

Arb Type	(1)	(2)	(3)	(4)	(5)	(6)
Number:	0	0	152	6	6	0

which is 0.664% of the overall generated variates. The main source of arbitrage stems from butterfly arbitrage and it is found at the shortest maturity for the lowest strike. Thus, the volatility arbitrage repair has only to be applied very rarely.

From our test, we observed that if the training data is arbitrage free the resulting GMM generated data is nearly arbitrage free. This means that a cross validated model captures the observed joint probabilities in the training data without overfitting.

Dynamic Arbitrage and $\mathbb{Q}$ Measure

This article is about synthetic market data generation using the $\mathbb{P}$ measure. We used GMM to approximate the joint distribution of given financial objects and their relationship and their timely evolution. The *dynamic arbitrage* references to the timely evolution and refers to the absence of a trading strategy that earns riskless profits. We described ways to guarantee that each generated object is arbitrage free but the timely evolution may not! For instance we modelled the 30 day yield curve dynamic as a multi-dimensional GMM. This involves that the resulting curves at $t, t > 1$, depend on several curves observed before t . This leads to a non-Markovian modelling and may not be compatible with the dynamic applying the risk-neutral measure, $\mathbb{Q}$.

If we wish to consider derivatives pricing, modelling arbitrage free timely evolution under $\mathbb{Q}$ is essential. Mathematically, this problem can be tackled using partial differential equations. For an n -dimensional stochastic process $x(t)$ with generic i -th component following an stochastic differential equation with W_t being a correlated n -dimensional Brownian motion, given by

$$\frac{dx_i(t)}{x_i(t)} = \mu_i(t)dt + C_i(x, t)dW_t,$$

μ_t and $C_i(x, t)$ deterministic functions of time and state, the associated *Kolomogorov forward PDE* - assuming proper initial and boundary conditions - for the density is

$$\frac{\partial p_t}{\partial t} + \sum_{i=1}^n \frac{\partial}{\partial x_i} [\mu_i(t)x_i p_t] - \frac{1}{2} \sum_{i,j=1}^n \frac{\partial^2}{\partial x_i \partial x_j} [C_i C_j x_i x_j p_t] = 0. \quad (10)$$

A solution to consider such dynamics are outlined in [3], where a multi-dimensional local volatility model based on mixtures of (log-)normal distributions is constructed. However, the calibration is done differently to the approach considered here. We can easily accommodate their set-up into our modelling as follows: For the mixtures we consider a covariance structure that only fits the marginal distributions and applying their calibration algorithm by selecting the parameters of the mixtures such that observed option prices are matched adequately.

Another approach would be, to start from the observed GPCs of a fitted GMM and assign a dynamic to these factors. For instance, if we have K GPCs for a yield curve evolution, we take each GPC as being the initial yield curve. Then, we use this as an input to a term structure model. The resulting model is a term structure mixture model with the weights resulting from the GMM fit. The remaining parameters are calibrated to observed option prices. This allows to take into account observed shapes from the yield curve history. Using such an approach, a low-parametric term structure model may suffice to model complex behaviour like skews and smiles. More complex models have a large number of parameters and it may be unstable to fit a model consisting of K such models. Another aspect that is worth to mention is that the resulting model is not grounded in the class of GMM anymore.

Remark: Using our approach it would be possible to fit a GMM to a spot value, a forward curve and implied volatility surface. Using conditioning it would then be possible to generate an implied volatility surface given a spot and a forward curve. Furthermore, combining this for different markets and, or a term structure approach is also feasible. Using such complex methodologies increase the runtime for fitting the GMM and also for the simulation according to the number of factors.

Conclusions, Summary and Future Research

We have proposed a framework based on Gaussian Mixture Models that is capable of generating market data that closely resembles the observed joint distributions. This includes multi-dimensional and conditional settings as well as dynamics that rely on n observed realizations. Non-Markovian settings can be considered and used for scenario generation. The stochastic sampling involves only the Gaussian and the uniform distribution. Application to times series generation for assets, curves and surfaces. Applications as backcasting, conditional sampling and generating illiquid data is considered and outlined using examples. The use of an objective criterion for backcasting and nowcasting is proposed. Further applications, for instance, forecasting opening prices for assets on exchanges that open at different times learning the conditional distributional relation of opening prices can be considered.

We have demonstrated that the method is versatile and only needs a reasonable amount of market data. In this respect it can be seen as an key advantage to methods relying on neural networks, like Variational Autoencoders or Generative Adversarial networks. It can also help to produce training data for applying such methods. Interpretability is another advantage of the approach. Using the Gaussian Principal components - the means of the constituting distributions - and the weights the movement and the importance of principal shapes is possible.

Finally, we considered the notion of no-arbitrage and gave an outlook to applying the method to include the dynamic in $\mathbb{Q}$ measure. This and linking the $\mathbb{P}$ and $\mathbb{Q}$ measures is ongoing research and is explored to be further in the future.

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